

EarthPress design precis

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DOCUMENT CONTROL

Document ID	EPR-PRC-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/02-concept.md @ ab4786d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

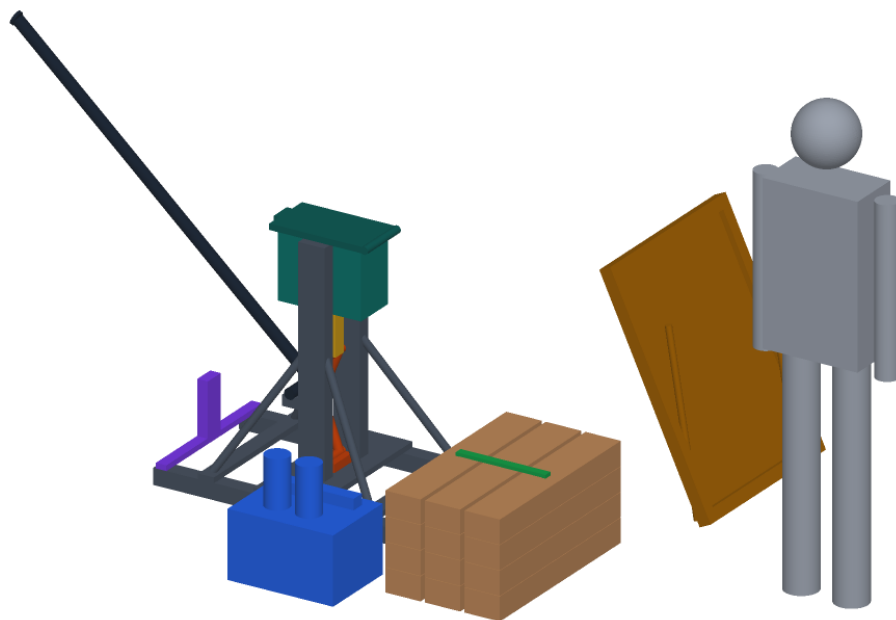
EarthPress design precis

■ Summary

EarthPress is a manual press, welded from common steel sections, that compacts moist, sieved and cement-stabilized site soil into 290 x 140 x 90 mm blocks with a single lever driving a toggle linkage under the mold. The same lever ejects the block once the lid swings open. A field soil test kit, a 5 mm sieve and a block gauge complete the set. First-order estimates suggest that about 81 kN on the block face gives the 2 MPa compaction pressure used in CSEB practice, that a crew of four could make about 300 blocks a day (about 9 m² of 140 mm wall), and that the press and kit cost about \$403 in parts. Two requirements are at risk and one is not met: the work needed per block may exceed one operator's stroke in the modeled layout (R2), the lever ends too low (R5), and the frame is heavier than the 50 kg single-piece limit (R7). All values are estimates to be checked at TRL 3.

EarthPress

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. EarthPress with its soil sieve, test kit, a stack of blocks with the block gauge, and a 1.75 m person for scale. Concept massing model, not for fabrication.

■ How it works

1. **Test.** The soil test kit screens site soil with a jar sedimentation test (sand, silt and clay fractions), a shrinkage box (linear shrinkage), and ribbon and drop tests. A printed chart turns the results into go or no-go and a starting cement content, usually 5 % by dry mass, following Auroville Earth Institute practice ([Auroville Earth Institute](#)).
2. **Sieve and mix.** The crew throws dry soil through a 5 mm mesh sieve leaning on a prop, measures soil and cement by bucket, mixes them dry, then adds about 10 % water until a squeezed ball holds its shape and breaks cleanly when dropped.
3. **Fill.** With the lever up and the lid open, the piston sits at the bottom of its stroke. The operator fills the 290 x 140 mm mold with a measured scoop of loose mix, about 150 mm deep, strikes it level and closes and latches the lid.
4. **Press.** The operator pulls the 1.5 m lever down. A short crank on the lever pushes the knee of a toggle linkage under the piston toward straight. The piston rises about 60 mm, compacting the soil to about 90 mm. As the toggle straightens its force ratio climbs steeply, so the peak force arrives at the end of the stroke, where the soil is stiffest.
5. **Eject.** The operator returns the lever, unlatches and swings the lid clear, then pulls the lever again past the compaction position to the ejection stop, which lifts the piston flush with the mold rim. The block is lifted off, checked with the block gauge and stacked in shade.
6. **Cure.** Blocks are kept damp under cover for about four weeks to let the cement hydrate, then air dried before use.

EarthPress: cutaway

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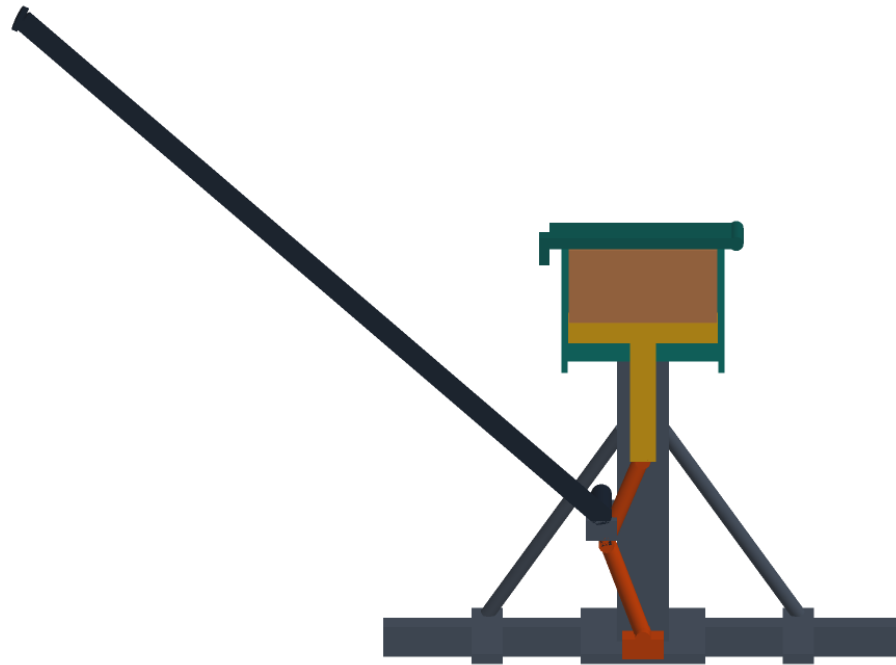


Figure 2. Section through the mold and linkage at the start of the stroke: loose soil (brown) above the piston (yellow), toggle links (orange) under the push rod, lever and crank on the left.

EarthPress: material flow per 100 blocks, kg (estimates; mixing adds 33 kg cement and 70 kg water)

CONCEPT, NOT FOR FABRICATION



Figure 3. Material flow per 100 blocks, in kg. All values are estimates for a soil with 15 % oversize, 5 % cement and 10 % water.

■ Main components

Numbers match the exploded view (Figure 4) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	Frame on skids	Two 100 x 50 mm channel columns, 60 x 60 mm tube skids about 1 m long, 20 mm base plate, round-bar braces	Carries the full block force as tension between the base pivot and the lid
2	Mold box	12 mm plate welded around a 290 x 140 x 250 mm cavity, with replaceable 3 mm wear liners (proposed)	Bolted to the columns so it can be removed (proposed, see R7)
3	Lid with hinge and latch	22 mm plate on a 28 mm hinge pin, hook latch on the lever side	Reacts the block force; must never open under load
4	Piston and push rod	20 mm plate on a 50 x 50 mm bar	Clearance about 1 mm per side in the mold
5	Toggle linkage and pins	Paired 20 mm flat links, about 190 mm between centers; three 30 mm pins in bronze or steel bushings	Knee moves about 100 mm sideways for about 60 mm of piston travel
6	Lever and crank	1.5 m schedule 40 pipe (about 42 mm OD) on a crank about 80 mm long; removable	Lever ratio about 19:1
7	Ejection stop and lever catch	Adjustable stop on the column; catch on the skid that holds the lever at rest	Stops the lever from falling when released
8	Soil sieve	5 mm welded mesh in a 700 x 900 mm timber frame on a prop	Removes stones and clods
9	Soil test kit	Two 1 L clear jars, a 600 x 40 x 40 mm shrinkage box, measuring buckets, spring scale, printed chart	Field tests only; no laboratory
10	Block gauge	Go or no-go bar for length and height	Checks every block

EarthPress: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Frame on skids
- 2 Mold box, 290 x 140 mm cavity
- 3 Lid with hinge and latch
- 4 Piston and push rod
- 5 Toggle linkage and pins
- 6 Lever and crank, 1.5 m
- 7 Ejection stop and lever catch
- 8 Soil sieve, 5 mm mesh
- 9 Soil test kit
- 10 Block gauge

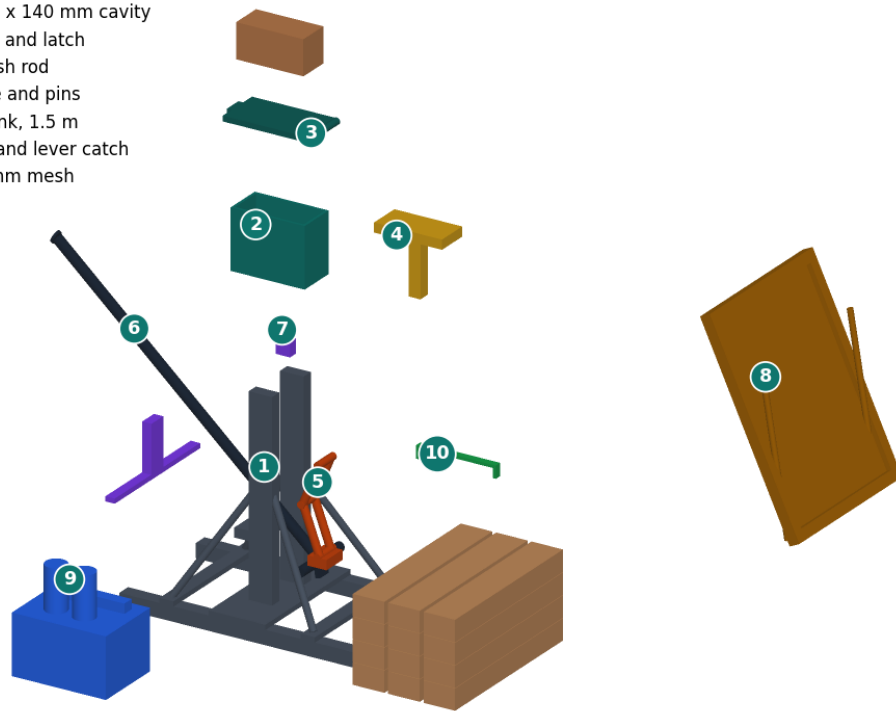


Figure 4. Exploded view with callouts matching the BOM. Loose soil and the block stack are context only.

■ First-order numbers

All values are estimates with the assumptions stated.

Quantity	Estimate	Basis
Block face area	0.0406 m ²	290 x 140 mm
Block force for 2 MPa	about 81 kN (8.3 tonne-force)	2 MPa x 0.0406 m ² ; Auroville practice is 2 to 4 MPa
Block force for 4 MPa	about 162 kN	Out of reach for a manual toggle press of this size
Overall force ratio needed	about 162:1	81 kN / 500 N pull at the grip
Lever ratio	about 19:1	1.5 m lever on an 80 mm crank
Toggle ratio needed at end of stroke	about 9:1	Links within about 3° of straight (ratio about 1 / (2 tan θ))
Compaction work per block	about 0.6 to 1.2 kJ (estimate)	Peak force x 60 mm stroke / 4 to / 8, since pressure rises steeply near the end
Work from one operator	about 0.65 kJ	500 N over about 1.3 m of grip travel (50° arc). Short of the upper estimate: a 100° arc or two operators needed
Block volume and mass	about 3.65 L, about 6.9 kg dry	Dry density about 1,900 kg/m ³ (assumed)
Cement per block	about 0.33 kg	5 % of dry soil mass
Output	about 300 blocks per 8 h day, crew of four	About 96 s per block for fill, press, eject and stack (assumed, to be timed)
Wall per day	about 9 m ² of 140 mm wall	About 33 blocks per m ² with 10 mm joints
Small house, about 50 m ² of wall	about 1,700 blocks, about 6 press days, about 560 kg cement (11 bags of 50 kg)	22 m perimeter, 2.7 m walls, 15 % openings
Embodied CO ₂ of that wall material	about 0.35 t for CSEB against about 4.6 t for fired brick	7.1 m ³ at 49 and 643 kg CO ₂ /m ³ (Auroville Earth Institute); excludes mortar and transport
Pin shear, 30 mm pin in double shear	about 57 MPa at 81 kN	Well within mild steel; bearing and link buckling to be checked at TRL 3
Column tension	about 40 MPa at 81 kN	Two channels, about 1,000 mm ² each (assumed)
Press mass	about 123 kg; frame about 61 kg, mold about 23 kg, lid about 11 kg	Steel at 7,850 kg/m ³ over nominal sections
Parts cost	about \$403	bom/bom.csv , indicative

■ Key design choices

All of these are proposed, awaiting Amish. The review note (docs/REVIEW.md) gives options and a recommendation for each.

- **Bottom piston with a toggle, one lever for both pressing and ejection**, following the CINVA-Ram pattern rather than a screw, a bottle jack or a top-down ram. It needs no hydraulic parts and a welder can build and repair it.

- **Flat 290 x 140 x 90 mm block**, the module used by the Auroville Earth Institute, rather than a larger or interlocking block. Interlocking molds could follow as inserts.
- **2 MPa compaction target**, the low end of CSEB practice, because 4 MPa needs about 162 kN, beyond a manual press of this size.
- **Cement stabilization at 5 % by default**, with lime as an option for clay-rich soils.
- **Field soil tests only**, with no lab equipment, so the kit costs little and travels with the press.
- **Bolted mold and removable lever**, so the press splits into pieces two people can lift (R7 is not yet met).

■ Safety

Safety: The press applies about 81 kN (over 8 tonne-force) at the piston. Keep hands out of the mold and away from the linkage while anyone is on the lever. Only the operator touches the lever, and only after calling "clear".

Safety: The lever stores energy near the end of the stroke and can kick back if the grip slips or a pin fails. Use the lever catch whenever the lever is released, keep the head out of the lever arc and never stand on the lever.

Safety: Never unlatch the lid while the lever is loaded. A lid or latch failure under load can throw the lid and soil upward. The latch must hold at least 1.5 times the peak block force (R12).

Safety: Portland cement is alkaline and can burn skin and eyes. Wear gloves, long sleeves and eye protection when mixing. Sieving dry soil raises dust that may contain respirable crystalline silica; sieve damp soil where possible, work upwind and wear an FFP2 or N95 respirator.

Safety: A dry block weighs about 7 kg and the press about 123 kg. Lift blocks close to the body, rotate tasks in the crew, and move the press with four people or dismantled.

Safety: Blocks made by EarthPress are not certified. Walls in seismic zones, multi-storey walls and any building that needs a permit require a structural design and block tests by a qualified engineer under the local building code.

Safety: Building the press involves welding, grinding and cutting. Use welding PPE, guard grinders and work away from flammables.

■ Open questions

- [] Raise the lever pivot to about 800 mm to give a 100° arc and a higher grip at the end of the stroke? This would help both R2 and R5.
- [] Bushings: bronze, steel or UHMW-PE for dusty conditions?
- [] Does the mold need hardened wear liners, or is plain 12 mm plate enough for 50,000 blocks with sandy soil?
- [] How do fill-mass errors change final pressure and block height with a toggle this close to straight? A variable-height stop may be needed.
- [] Is lime stabilization practical where cement is scarce, and does the press need a longer dwell for it?
- [] Which partner and region should the first co-design sessions use?